K-map

Simplification

3, 4 variables

+ 01 :

SAMARTH

CUET PG 2027

Lecture – 11 | Computer Science

Unit 03 : Digital Fundamental

Topic : Kmap Simplification: Part 2

By- Puneet Kaur



Recap *of previous lecture*

1. Kmap Simplification: Part 1



Topics *to be covered*

1. Kmap Simplification: Part 2



Topic : Kmap Simplification: Part 2

Topic : Kmap Simplification: Part 2

3- Variable K- Map:

- A Karnaugh map in three variables is a rectangle divided into eight squares, One of the ways that eight possible mintena are labeled in squares. Show the alternate way of representing three variables. In addition to square which are physically adjacent, leftmost and rightmost columns of K-map differ in only one variable. Thus A'B'C' and AB'C' adjacent, and so are A'B'C and AB'C.

Topic : Kmap Simplification: Part 2

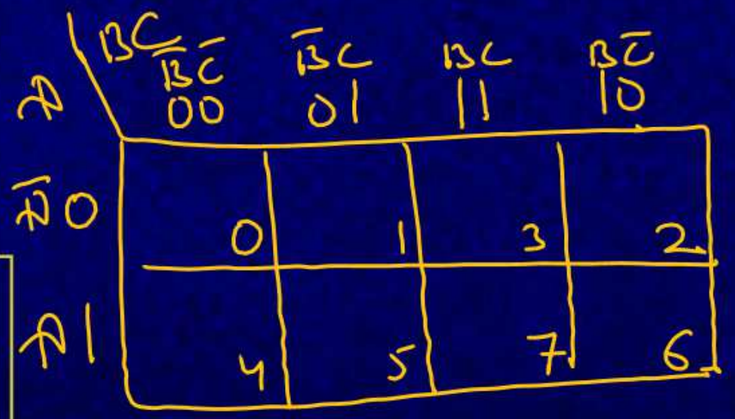
	A'B'	A'B	AB	AB'
C'	A'B'C'	A'BC'	ABC'	AB'C'
C	A'B'C	A'BC	ABC	AB'C

C \ AB	00	01	11	10
0	A'B'C'	A'BC'	ABC'	AB'C'
1	A'B'C	A'BC	ABC	AB'C

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- Given a minterm expansion of a function, it can be plotted on a map by placing Is in the Squares which corresponds to minterms present in the expressions and 0s in the remaining squares. Kamaugh map of the Boolean expression A'B'C' + A'B'C + A'BC' + ABC'.

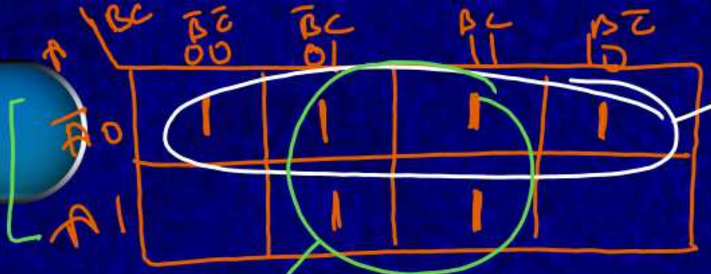
C \ AB	00	01	11	10
0	1	1	1	0
1	1	0	0	0



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- To simplify a sum-of-products expansion in three variables, one has to identify groups of terms that can be combined. While forming groups of squares containing Is the following considerations must be kept in mind.
  (i) The number of squares in a group must be equal to 2" such that 2, 4, 8, 16.
  (ii) A square containing can be included in as many groups as desired.
  (iii) Group must be largest possible groups, a group of two squares containing 1 should not be made if these squares can be included in a group of four squares. A K-map that contains group of four Is that are adjacent to each other is called a quad. Looping a quad of Is eliminates the two variables that appear in both complemented and un-complemented form.

Topic : Kmap Simplification: Part 2



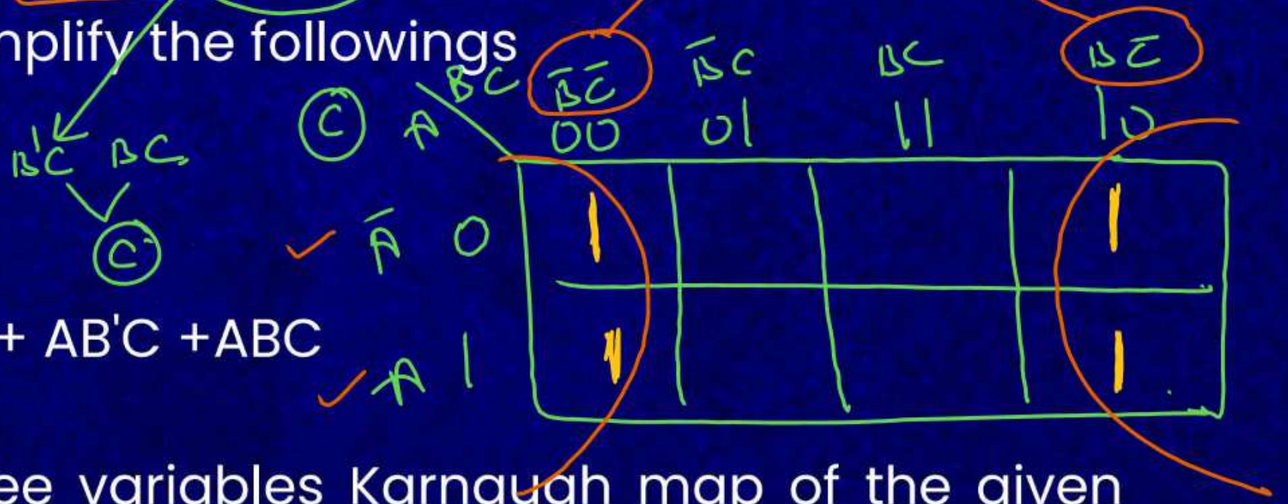
Example: Use Karnaugh map to simplify the followings

(a) X = ABC'+ ABC

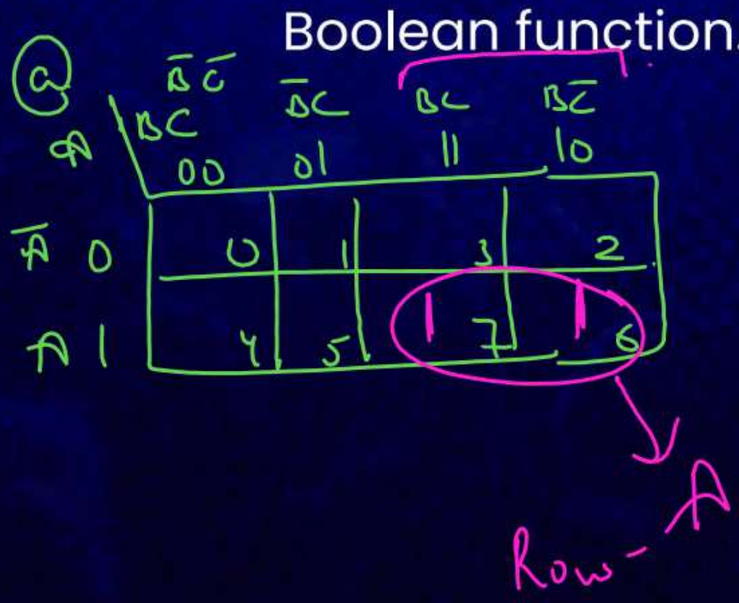
(b) X = A'B'C'+ AB'C'

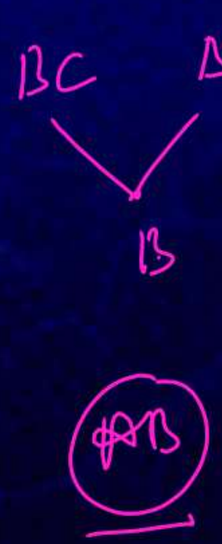
(c) X = A'B'C'+ A'BC' + ABC' + AB'C'

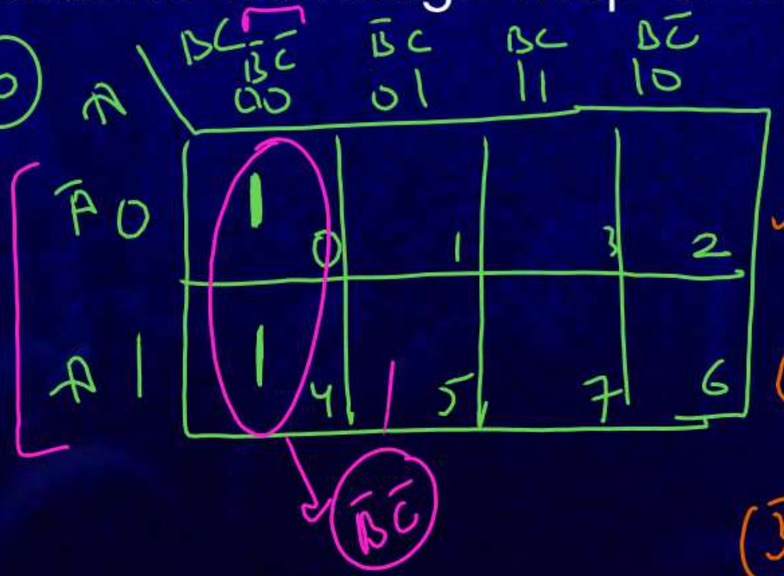
(d) X = A'B'C'+ A'B'C + A'BC + A'BC' + AB'C +ABC



Solution: (a) it represents three variables Karnaugh map of the given Boolean function.







Pair Aim

1. Quad (Pair of 4 1's)
2. Double (Pair of 2 1's)
3. Single.

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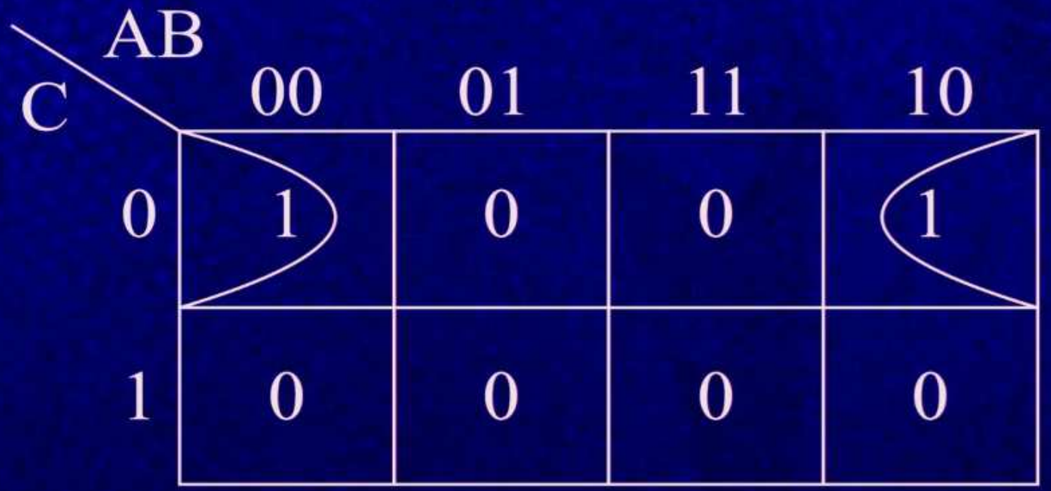
C \ AB	00	01	11	10
0	1	1	1	0
1	1	0	0	0

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- The adjacent squares representing ABC' and ABC are grouped together. This eliminates the C variable since it appears in both un-complemented and complemented form. The simplification function will be X=AB.

(b) The Boolean function X = A'B'C' +AB'C'. In a K-map, the left and right column of squares are considered to be adjacent. Then, the two 1s in this map can be looped to provide a simplified result X= B'C' (variable A is eliminated).

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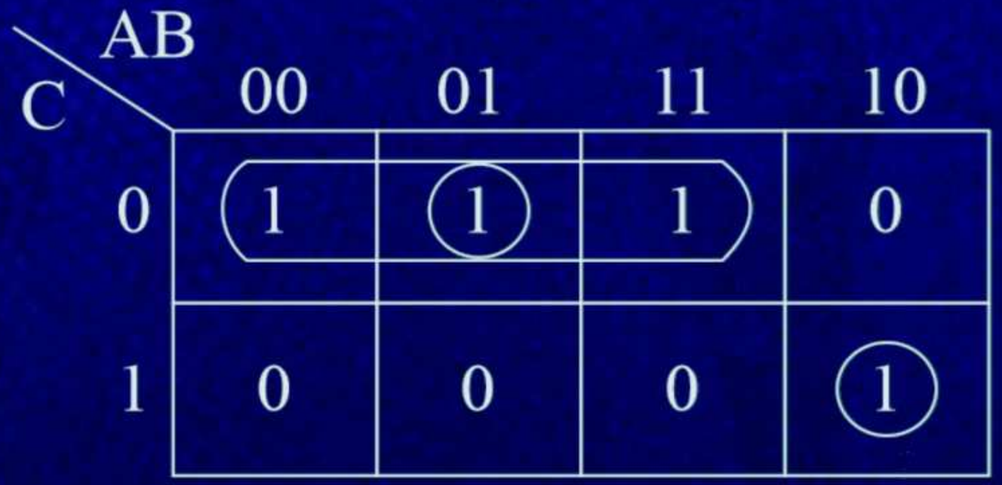


Topic : Kmap Simplification: Part 2

(C) It represents three variables Karnaugh map of the given Boolean function There are two pairs of Is and they can be combined. Only I representing the term AB'C is isolated. The group of first two horizontal I square gives A'C' and the group second and third horizontal I square gives BC'. Hence, the simplified result is

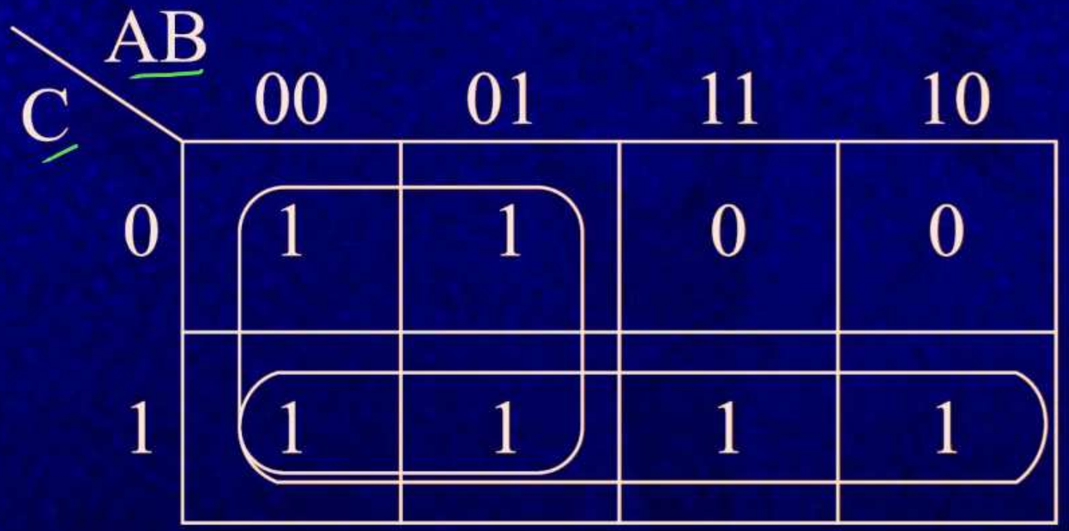
$$X = A'C' + BC' + AB'C$$

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(d) The Kamaugh map of the given function



Here two quads have been formed. Notice that 1s are a part of both the quads. The qand formed by A'B'C', A'B'C', A'BC' and A'BC produces the resultant as A' and the quad formed by A'B'C, A'BC, ABC and AB'C produces the resultant as C. Hence the final resultant expression is X= A' +C.

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4- Variable K- Map:

- A Karnaugh map in four variables is a square divided into 16(=24) squares. The squares represent the 16 possible minterms in four variables. The definition of adjacent squares to be extended so that not only are leftmost and rightmost column adjacent as in the 3-variable map, but also the first and last rows are adjacent (a) shows Karnaugh map for four variables A, B, C and D and (b) shows the alternate way of representing four variable. Note that A'B'C'D' square in the first row is adjacent to the square A'B'C'D' in the last row, since they differ only in the variable C. One can think of the top of the map as being wrapped around to touch the bottom of the map.

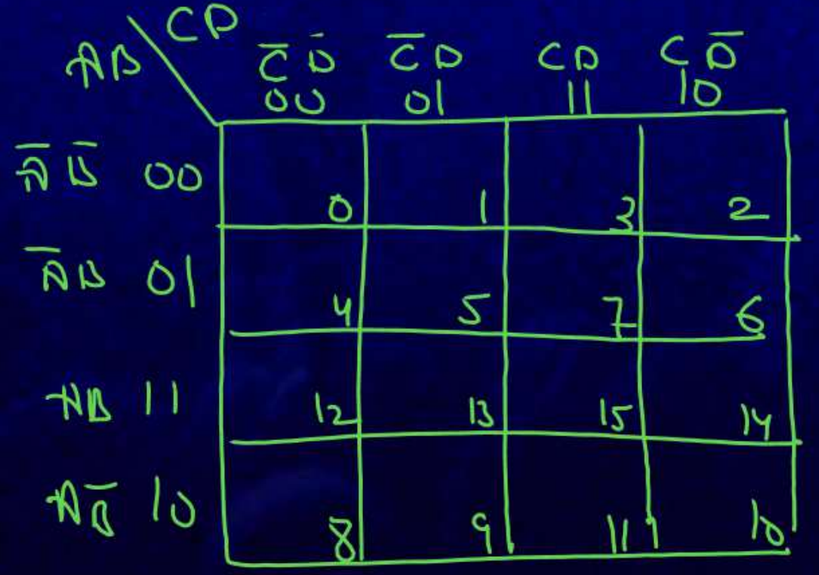
Topic : Kmap Simplification: Part 2

	A'B'	A'B	AB	AB'
CD'	A'B'CD'	A'BC'D'	ABC'D'	AB'C'D'
C'D	A'B'C'D	A'BC'D	ABC'D	AB'C'D
CD	A'B'CD	A'BCD	ABCD	AB'CD
CD'	A'B'CD'	A'BCD'	ABCD'	AB'CD'

CD \ AB	00	01	11	10
00	A'B'CD'	A'BC'D'	ABC'D'	AB'C'D'
01	A'B'C'D	A'BC'D	ABC'D	AB'C'D
11	A'B'CD	A'BCD	ABCD	AB'CD
10	A'B'CD'	A'BCD'	ABCD'	AB'CD'

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- To simplify a sum-of-product expansion in four variables, one has to identify group of minterms of squares 2 4, 8 or 16 containing 1s that can be combined. A group of eight 1s that are adjacent to one another is called an octet. Looping an octet of 1s eliminates three variables that appear in both complemented and uncomplemented form several examples of octet are:



Pair Quiz

(1) Octet (Pair of 8 1's)

(2) Quad (Pair of 4 1's)

(3) Double (Pair of 2 1's)

(4) Single

Topic : Kmap Simplification: Part 2

Quiet

CD \ AB	00	01	11	10
00	0	0	0	0
01	1	1	1	1
11	1	1	1	1
10	0	0	0	0

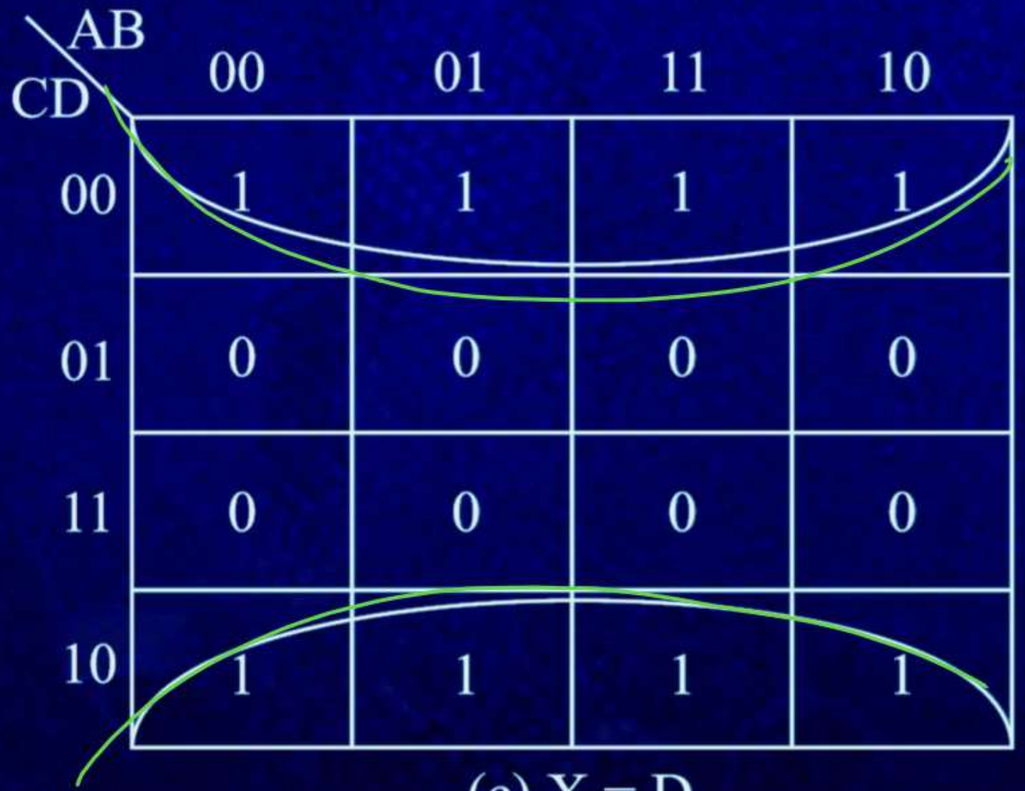
(a) X = D

Consecutive Rows & Columns can form Pair

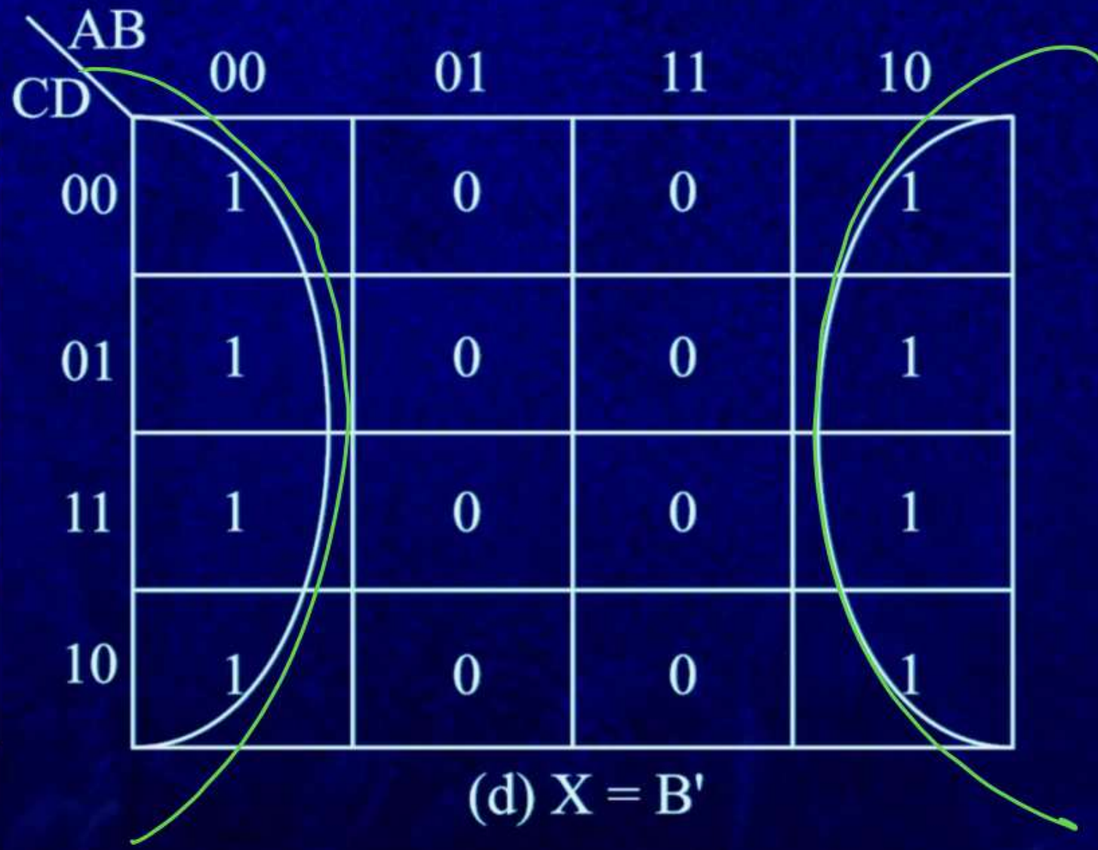
CD \ AB	00	01	11	10
00	1	1	0	0
01	1	1	0	0
11	1	1	0	0
10	1	1	0	0

(b) X = A'

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(c) X = D



(d) X = B'

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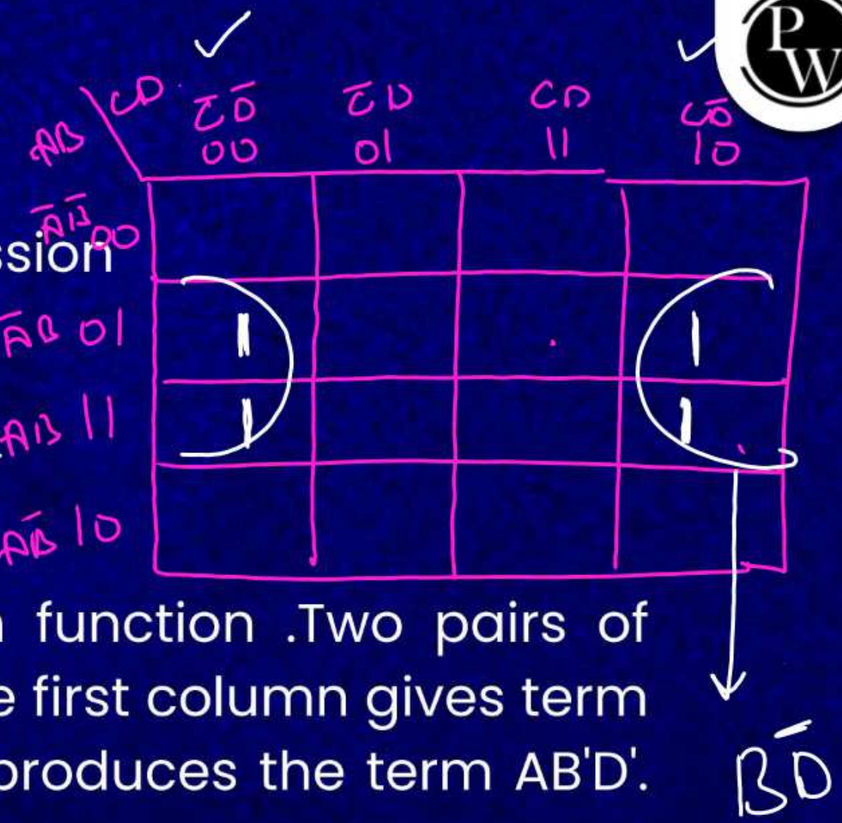
Example: Use K- map to simplify the following expression

(a) X= A′B′CD + A′B′CD′ +AB′C′D′ + AB′CD′

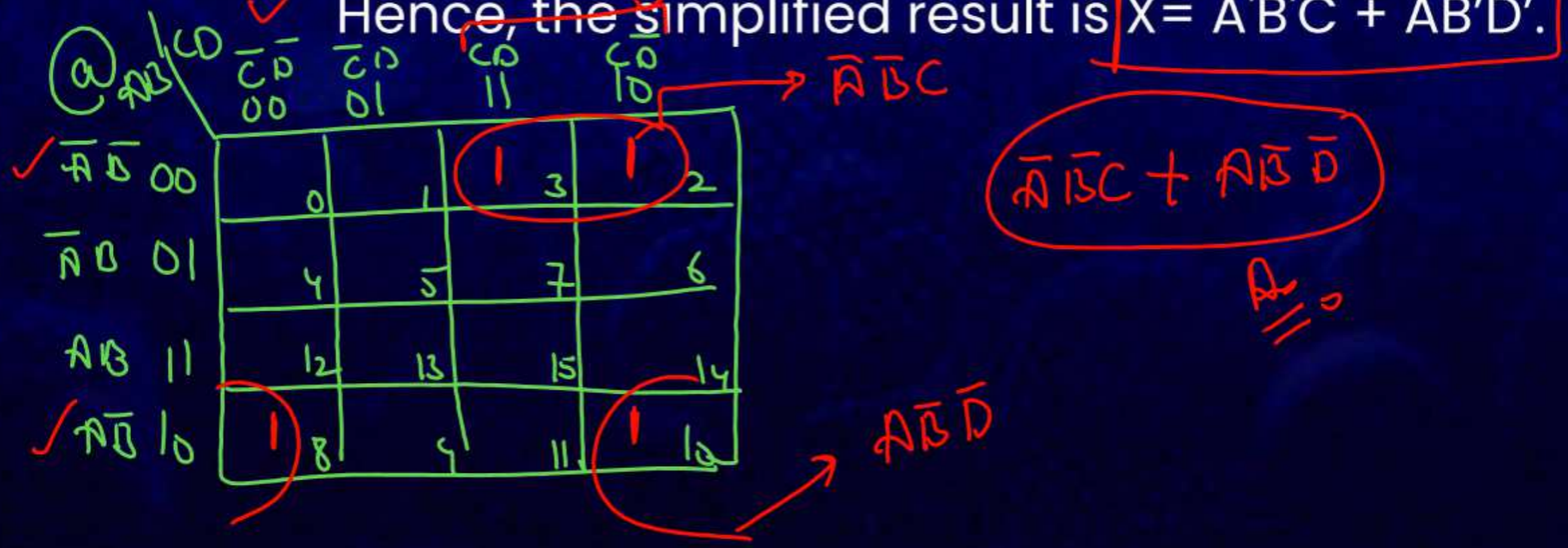
(b) X=A′BC′D′ +ABC′D′ +A′BCD′ +ABCD′

(c) X= A′B′C′D′ +AB′C′D′ +A′B′CD′ +AB′CD′

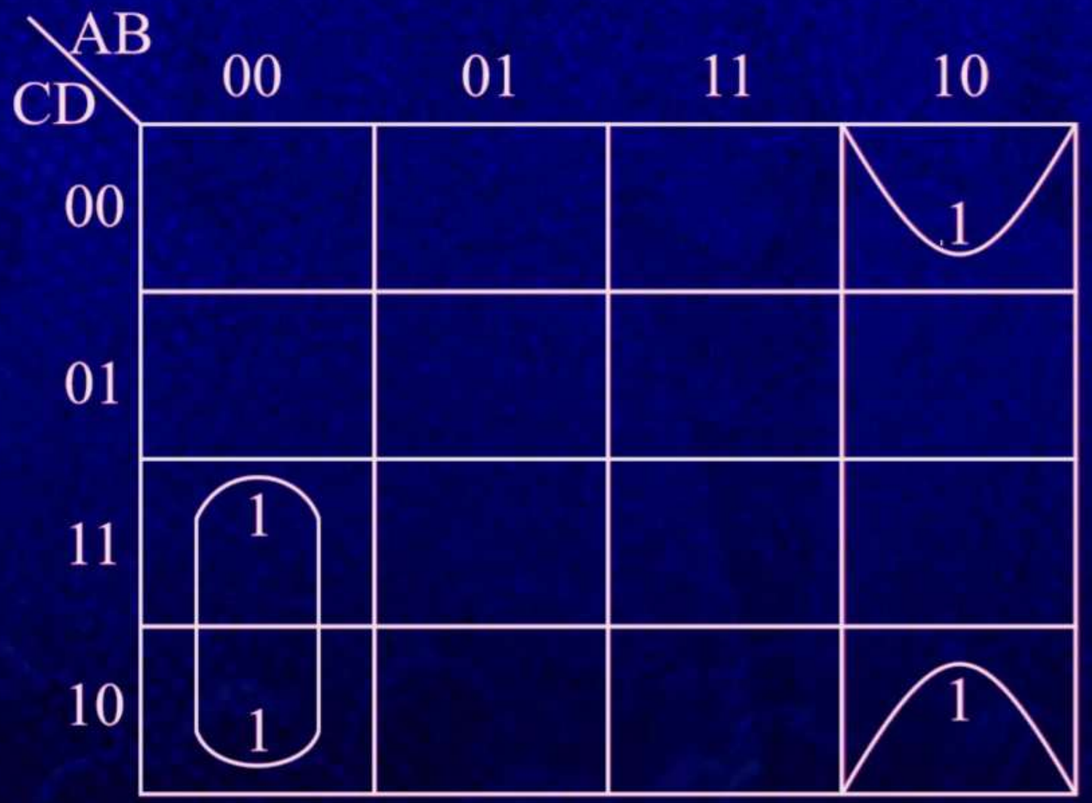
H.W



Solution: (a) The Karnaugh map of the given function .Two pairs of adjacent Is are grouped together. The pair in the first column gives term A'B'C and the pair of top row and bottom row produces the term AB'D'. Hence, the simplified result is X= A'B'C + AB′D′.



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Topic : Kmap Simplification: Part 2

(b) The Karnaugh tip of the given Boolean function. Two 1-Squares in the first row and two 1-square in the last row f are adjacent squares and hence they form a quad. The variables B and D' remain unchanged (A and C being complemented and uncompleted form). The resultant expression is X=B.D'.

SOP = Σ

POS = Π

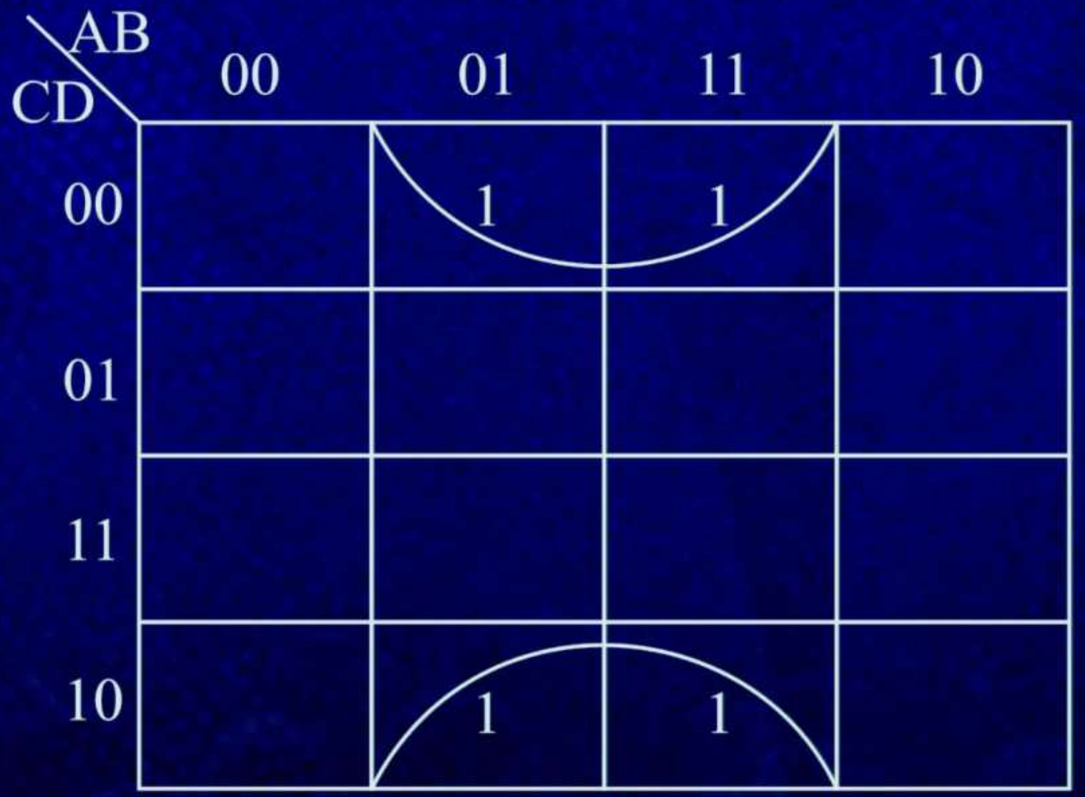
Q. $F(ABCD) = \Sigma(0,1,2,3,4,5,6,7,8,9,11)$

Simplify using k-map.



$\bar{A} + \bar{B}\bar{C} + \bar{B}D$

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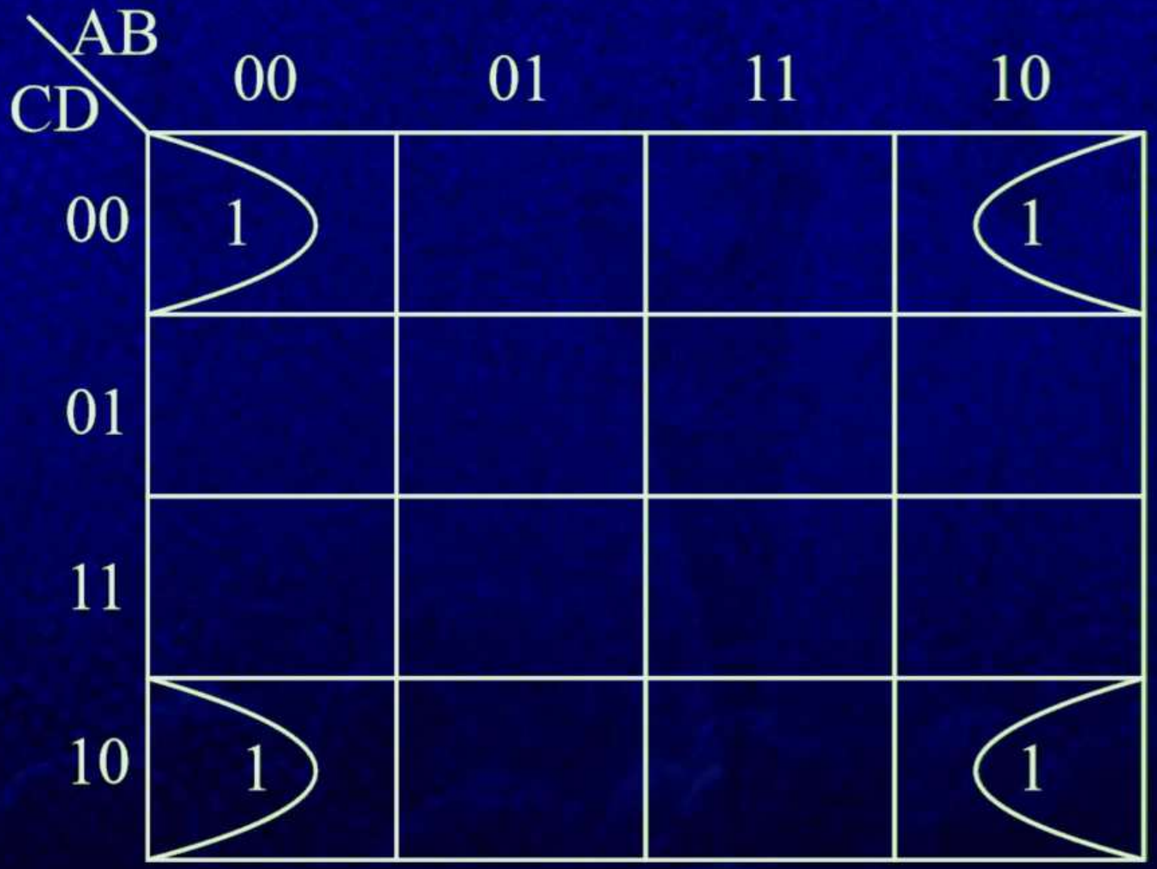


Topic : Kmap Simplification: Part 2

(C) The Karnaugh map of the given Boolean function. Four squares are adjacent squares, the top and bottom rows are considered to be adjacent to each other as are the leftmost and rightmost columns. The variables B′ and D′ remain unchanged (A and Care in complemented and uncomplemented form).

The resultant expression is X = B'.D′.

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Topic : Kmap Simplification: Part 2

Complete Simplification Process-

- We have seen that on a K-map, a loop of two eliminates one variable, a loop of four eliminates two variables, are a loop of eight eliminates four variables. Therefore, to eliminate maximum number of variables, one must form the grouping with maximum number of consecutive terms having marked I. However, number of such consecutive terms, forming a group, must be expressed in powers of 2. The steps below are grouped in using the K-map method for simplifying a Boolean expression.

Topic : Kmap Simplification: Part 2

1. Construct the Karnaugh map and place in those squares corresponding to the minterm present in the expression and 0s in other squares
2. Find out the squares containing Is which cannot be grouped with any other square. Write the minterms corresponding to those squares.
3. Look for those squares containing 1s which are adjacent to only one square containing 1s. Loop for any pair containing such a 1 and write the simplified Corresponding terms.

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4. Loop any octet even it contains some 1s that have already been looped and write the simplified corresponding terms.
5. Loop any quad that contains one or more 1s which have not been looped, making sure to use the minimum number of loops.
6. The simplified expression of the given Boolean expression is the sum of all the terms derived during the process from (2) to (4).

Topic : Kmap Simplification: Part 2

Double negation law

$(A')' = A$

A	A'	(A')'
0	1	0
1	0	1

#Q. Find suitable option for the given k-map.

RS \ PQ	$\bar{P}\bar{Q}$ 00	$\bar{P}Q$ 01	PQ 11	$P\bar{Q}$ 10
$\bar{R}\bar{S}$ 00	0	1	1	0
$\bar{R}S$ 01	1	1	1	1
RS 11	1	1	1	1
$R\bar{S}$ 10	0	0	0	0

$\bar{R}Q$

S

$R\bar{S}$ $\quad \bar{S}\bar{Q}$

SOP = $R\bar{S} + \bar{S}\bar{Q}$

↓ Complement

A. $Q\bar{R} + S$
B. $QR + S$
C. $Q\bar{R} + \bar{S}$
D. $QR + \bar{S}$

$(R\bar{S} + \bar{S}\bar{Q})'$

$(R\bar{S})' \cdot (\bar{S}\bar{Q})'$

$(\bar{R} + \bar{\bar{S}}) \cdot (\bar{\bar{S}} + \bar{\bar{Q}})$

$= (\bar{R} + S) \cdot (S + Q)$ → POS

$\bar{R}Q + S$

Terms

Octet	Has	1 variable
Quad	Has	2 variable
Double	Has	3 variable
Single	Has	4 variable

By default = SOP

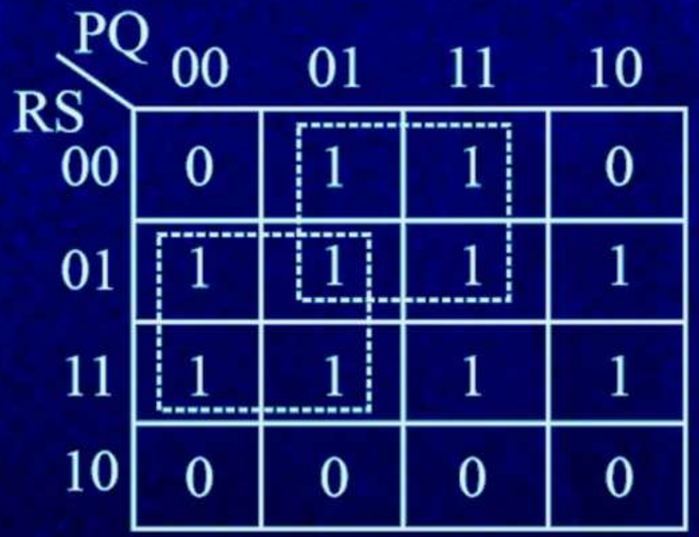
POS

- Consider 0's
- Find SOP of 0's
- Complement it

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Ans. A

From given k-map we have following pair formation



Thus, we have output = S + QR

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X — Don't care condition

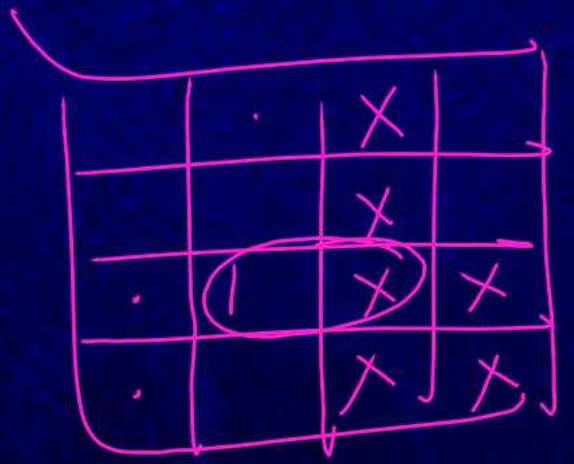
Consider X only if u have to create larger pair.

#Q. The minimized expression for the given K-map is

CD \ AB	00	01	11	10
00		1	x	
01			x	
11	1	1	x	x
10	1		x	x

$B\bar{C}\bar{D}$ CD $C\bar{B}$

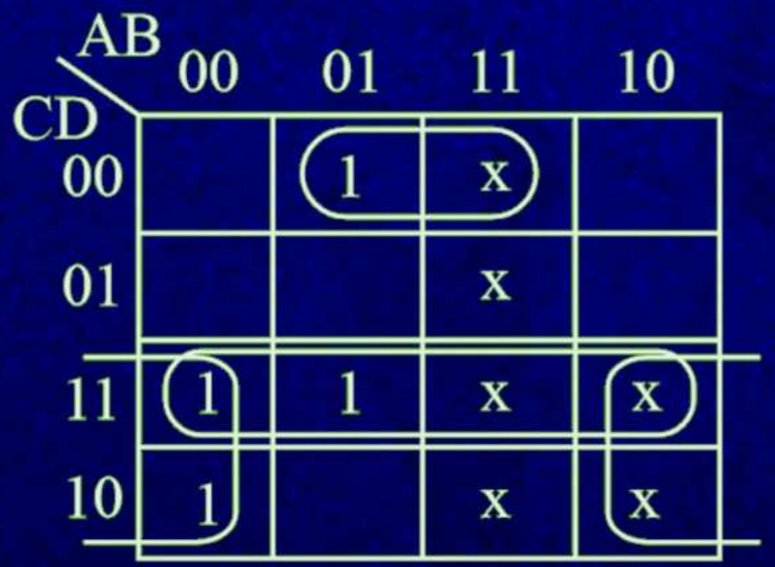
A. $\bar{A}B\bar{C}D + \bar{A}CD + \bar{A}\bar{B}C\bar{D}$
B. $\bar{A}B\bar{C}\bar{D} + \bar{A}CD + \bar{A}BC$
C. $B\bar{C}\bar{D} + CD + \bar{A}BC$
D. $B\bar{C}\bar{D} + CD + \bar{B}C$



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Ans. D

Using K-map:



Pairing, we get, f = $B\bar{C}\bar{D}$ +CD+ $B\bar{C}$

Hence, option D is correct.

2 Mins Summary

K-map simplification 3, 4 variables

+ DI's

Homework from SAMARTH CUET PG 2027 Module

Q. $F(ABCD) = \Sigma(0,1,2,8,9,12,13)$

$d(A,B,C,D) = \Sigma(10,11,14,15)$

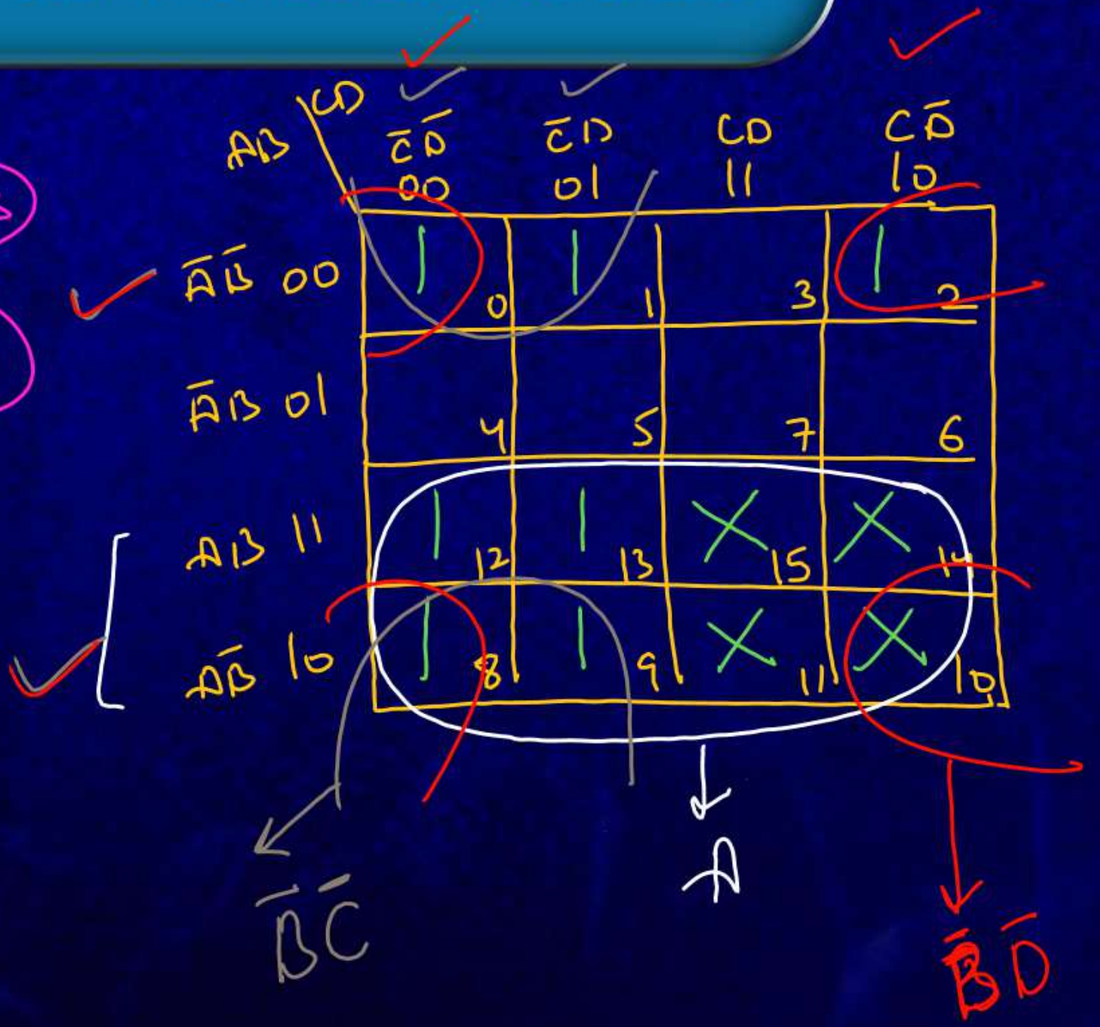
Simplify

(1) $A + \bar{B}\bar{D} + BC$

(2) $A + \bar{B}\bar{D} + \bar{B}\bar{C}$

(3) $\bar{A} + \bar{B}\bar{C}$

(4) $\bar{A} + \bar{B}C + \bar{B}\bar{D}$



THANK YOU